

A Comparative Study of CIPN Symptom Estimation Methods Based on Machine Learning

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ABSTRACT

Chemotherapy-induced peripheral neuropathy (CIPN) is a common side effect of anticancer medicines that causes muscular weakness and falls in cancer patients. Therefore, we compared the methods for identifying and estimating the factors that cause symptoms in CIPN based on factor analysis and four different machine learning algorithms. The results of factor analysis showed that the latent factors causing CIPN can be classified into three categories: “Morning body condition”, “Pain in the legs” and “Dizziness”. We chose models for machine learning based on the premise of multi-task learning and built numerous machine learning models to estimate various CIPN symptoms. According to the results of the comparison studies, TabNet has the best generalization performance.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Empirical studies in ubiquitous and mobile computing**; **Field studies**.

KEYWORDS

chemotherapy-induced peripheral neuropathy; fall detection; mobile sensing; machine learning

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1 INTRODUCTION

Cancer is a common disease. One in two Japanese people will suffer from cancer and cancer is the leading cause of death in Japan with 27.9%. However, recent advances in treatment technology and the development of highly effective anti-cancer drugs have led to the general use of anti-cancer drugs for the treatment of cancer.

Chemotherapy-induced peripheral neuropathy (CIPN) is a common side effect of cancer treatment medications [2]. The most common symptom of CIPN is numbness in the extremities due to peripheral nerve damage, which frequently leads to falls. This has a significant influence on both therapy and everyday living [6]. In a previous research, 40.9% of patients had falls as a result of CIPN [9]. Health-care costs associated with falls among cancer survivors in the U.S. have been considered to represent a danger to the health-care system [16]. Because of the increasing prevalence of outpatient cancer treatment, there is a need for preventative fall care that is closely connected with the lifestyles of patients [13].

Estimating the users' status of CIPN symptoms is critical for a fall prevention care system [12]. For example, Sasaki et al [12] proposed to estimate the users' status by using supervised machine learning. However, this research adopts accuracy for metrics and fails to consider the case of imbalance in the objective variables. In fact, in this data, multiple objective variables are unbalanced, and the performance of the model cannot be properly grasped when evaluated with accuracy. There are two reasons why the objective variable are unbalanced. First of all, since the frequency of falls in the lives is, in most cases, only a few times a day at most, the collected sensor data and labels will naturally have deviations. Second, because of the COVID-19 pandemic, it is difficult to collect a large amount of data nowadays due to the tight resources of medical institutions as well as various constraints for the research experiment in year 2020.

Therefore, this research aims to examine how to accurately estimate symptoms from such unbalanced data, by focusing on machine learning as the first step to building a fall prevention care system. In this research, our approach contains two major components, i.e., factor analysis and supervised machine learning. Factor analysis is used to conduct a search for symptom factors on the collected questionnaire data. The search for symptom factors can be divided into two parts: the search for latent factors using exploratory factor

analysis, and confirmatory factor analysis to determine the validity of the results. Next, we constructed several machine learning models based on the premise of multi-task learning with output for multiple objective variables from a single input of data, and conducted comparison experiments based on our own evaluation metrics. For the comparison of symptom estimation methods, we built four models after appropriate preprocessing of the data. As a result of model selection, we adopted LightGBM, TabNet, LSTM, and Transformer. We divided 73% of the total data into training data and 27% into test data by using Leave-One-Subject-Out cross validation. As a result of this study, we confirmed that TabNet is the model with the highest generalization performance on our test data.

The contributions of this research are as follows.

- (1) We analyzed the questionnaire data of CIPN patients using multivariate analysis methods to explore and interpret the symptom factors and to determine the validity of the analysis.
- (2) We constructed a model that estimates multiple symptoms caused by CIPN, assuming multi-task learning with multiple objective variables from a single input.

2 RELATED WORK

2.1 Mobile sensing

Smartphone is outfitted with a multitude of sensors and can capture a wide range of data. It is not an exaggeration to say that smartphones have become an important tool in our lives, with more than 95% of Japanese people owning one. Smartphone sensing is a research technique used to better comprehend human behaviors. These studies are being carried out on a variety of dimensions and contexts, in disciplines such as computer science, social science, and public health [10, 15]. Smartphone sensing frameworks are constantly being developed. Such frameworks have the potential to significantly reduce development and maintenance costs. AWARE framework [5] is an Android platform for collecting sensor data such as hardware-based (accelerometer, GPS, barometer, etc.), software-based (battery, screen, network, etc.), and human-based (experience sampling method (ESM)) data.

2.2 Fall detection

Existing fall detection systems are classified into three kinds depending on the deployed sensors [4], ambient sensors [11], and vision-based sensors [17], in current fall detection studies. Fall detection technology, in particular, employing mobile and wearable devices, is being intensively explored. Smartphones are equipped with a variety of sensors, including 3-axis accelerometers, gyroscopes, and high-performance microprocessors, making them an excellent platform for fall detection. Stefano et al. built a smartphone-based fall detection system for older persons that accounts for everyday activities that might lead to erroneous detection [1]. Sasaki et al. constructed system for fall-detection on top of both smartphone and smartwatch (e.g., AppleWatch) and collected sensor data and user's annotation based on the combined use of those devices [12].

Table 1: Sensor Information

Device	Sensor Name	Sensing Frequency
iPhone	Accelerometer	30Hz
iPhone	Activity Recognition	1Hz
iPhone	Barometer	10Hz
iPhone	Battery	1Hz
iPhone	Bluetooth	1Hz
iPhone	Gravity	30Hz
iPhone	Gyroscope	30Hz
iPhone	Location	0.3Hz
iPhone	Network	1Hz
iPhone	Screen	1Hz
iPhone	Wi-Fi	1Hz
iPhone	Weather	0.02Hz
Apple Watch	Accelerometer	10Hz
Apple Watch	Roll	10Hz
Apple Watch	Pitch	10Hz
Apple Watch	Yaw	10Hz
Apple Watch	Gravity	10Hz
Apple Watch	Rotation rate	10Hz
Apple Watch	User-Acceleration	10Hz
Apple Watch	Heart rate	1Hz
Apple Watch	Barometer	1Hz

3 IDENTIFICATION OF FACTORS CAUSING CIPN SYMPTOMS

3.1 Data set

In this research, we a dataset previously collected in our past research [12]. We collected the sensor data and annotation data from 8 CIPN patients. We asked each participant to use a smartphone (iPhone) and a smart watch (AppleWatch) that contained our data collection software and asked data collection for 2 weeks. For more detailed information on data collection, the readers are encouraged to refer to [12].

The data set contains the following sensor data from various types of smartphone sensors (shown in Table 1). The sensor data was collected from the phone and watch continuously throughout the 2-week study period. The data set also contains subjective annotation on the participant's individual daily status asked through questionnaire (shown in Table 2). The questionnaire was displayed on the participant's device every morning during the study period.

Table 2: User's Answered Status Questions

Question
1: Do you feel well this morning? (Yes/No)
2: Is your physical condition good today? (Yes/No)
3: Do you feel like you're busy today? (Yes/No)
4: Do you feel light-headed? (Yes/No)
5: Do you feel any pain? (Yes/No)
6: Do you have numbness in your hands? (Yes/No)
7: Do you have numbness in your legs? (Yes/No)
8: Do your legs feel heavy? (Yes/No)

Table 3: The Result of Exploratory Factor Analysis

Questions	Factor1	Factor2	Factor3	Commonality
Q1: Do you feel well this morning?	1.068	0.214	-0.256	0.995
Q3: Do you feel like you're busy today?	-0.843	-0.198	-0.225	0.797
Q2: Is your physical condition good today?	-0.770	0.107	-0.066	0.680
Q6: Do you have numbness in your hands?	0.451	-0.293	0.250	0.479
Q7: Do you have numbness in your legs?	0.432	1.030	-0.177	0.995
Q8: Do your legs feel heavy?	-0.197	0.861	0.031	0.873
Q5: Do you feel any pain?	-0.124	0.475	0.456	0.454
Q4: Do you feel light-headed?	0.129	-0.090	0.952	0.995
Factor contribution ratio	0.352	0.264	0.168	
Cumulative factor contribution ratio	0.352	0.616	0.783	

3.2 Exploratory factor analysis

Factor analysis is assumed to have latent factors behind the data to be analyzed. We used the collected questionnaire data to identify and model the factors that contribute to the development of CIPN symptoms.

First, the eigenvalues were calculated. The number of eigenvalues with a value greater than one was three. Based on the decay of the eigenvalues, a three-factor solution was considered appropriate. Next, a parallel analysis was conducted, and again a three-factor solution was suggested. Based on these results, we adopted the three-factor solution and conducted an exploratory factor analysis (maximum likelihood method, Promax rotation). The estimation results are shown in Table 3.

The first factor showed high factor loadings for the three items, "Do you feel well this morning?", "Do you have numbness in your hands?" and "Do you have numbness in your legs?". This factor was named as the "Morning body condition" factor. The second factor showed high factor loadings for the next three items, "Do you have numbness in your legs?", "Do your legs feel heavy?" and "Do you feel any pain?". This factor was named as the "Pain in the legs" factor. Finally, for the third factor, two items, "Do you feel any pain?" and "Do you feel light-headed?" showed high factor loadings. They were judged to be factors related to light-headed. This factor was named as the "Dizziness" factor. The correlations between the factors were about ± 0.2 , and no specific trend was found. The cumulative factor contribution ratio was 78.3%. Thus, we can assume that there are three latent factors in CIPN.

3.3 Confirmatory factor analysis

Based on the findings of the exploratory factor analysis, for the three factors: "morning body condition," "pain in the legs," and "dizziness," we conducted a confirmatory factor analysis under a model in which each observed variable was influenced only by one of the factors considered to be strongly related to it.

All factors were assumed to be correlated. As a result of the analysis, the goodness-of-fit indices were CFI = .978, TLI = .918, RMSEA = .113 (90% CI [.021, .203]), and SRMR = .149, indicating that the model fit the data well. Therefore, this result suggested that these factors are three major aspects of CIPN symptoms.

4 INVESTIGATION OF CIPN SYMPTOM ESTIMATION METHOD

Next, we constructed a machine learning model for estimating the symptoms of CIPN from the collected smartphone sensor data.

At first, we extracted the various features from the raw smartphone sensors for each daily time window. For example, for raw acceleration data, various features such as magnitude, mean, variance, correlation, covariance, skew, kurtosis, entropy, energy distance are extracted in daily time windows. Next, we constructed supervised machine learning with features extracted from sensor data as explanatory variables and self-reported CIPN symptom data as objective variables.

In constructing the model, we assumed multi-task learning with outputs belonging to multiple objective variables for a single input, and aimed to predict each item in the questionnaire data simultaneously as a precondition. For model selection, we adopted four methods: LightGBM, TabNet, LSTM, and Transformer.

Then, we conducted the 5-folded cross-validation for each machine learning model to verify the performance.

We split all data using Leave-One-Subject-Out cross validation and set the training data to 73%. We constructed a classifier that simultaneously estimated multiple symptoms in CIPN, and evaluated it using test data.

4.1 Model selection

We built multiple models by using the following four algorithms and compared the performance.

LightGBM [8] is one of the frameworks for gradient boosting based on decision tree algorithms, which grows the decision tree "leaf-wise" during the gradient boosting process.

TabNet [3] is a deep learning model for table data. One of the features of this algorithm is that it has explanatory potential by using Sequential Attention. Since it is known that it is generally difficult to calculate the importance of variables in prediction with conventional deep learning models, TabNet can be said to have the same level of interpretation as tree-based algorithm.

LSTM [7] is a recurrent neural network model. LSTM has a structure for storing information and three gates for adjusting the amount of data transport.

Transformer [14] is a deep learning model that is mainly used in the field of natural language processing. This model is characterized

by the fact that it does not use any RNN or CNN layers, but only Attention.

4.2 Experiments

4.2.1 Evaluation metrics. In the experiment, we introduced the mean value of the logarithmic loss function (logloss) as evaluation metrics. Using cross-entropy, the logloss quantifies the deviation between the predicted probability and the correct answer data. If the loss is near to zero, the constructed model is judged to be capable of making accurate predictions. In this study, each objective variable's logloss is computed, followed by the average.

$$\text{score} = -\frac{1}{M} \sum_{m=1}^M \frac{1}{N} \sum_{i=1}^N [y_{i,m} \log(\hat{y}_{i,m}) + (1 - y_{i,m}) \log(1 - \hat{y}_{i,m})]$$

The reason for adopting this metrics is that it is difficult to improve metrics like accuracy and AUC at the same time when using multi-task learning to predict several columns simultaneously. This is due to the higher training time and computing cost imposed by the model's increasing complexity. Therefore, we determined that the logloss would be a useful metric for reducing the overall model's prediction error.

4.2.2 Experimental results. The results of the experiments using the methods above are shown in the Figure 4. The model with the best generalization performance on the test data was found to be TabNet. The individual considerations for each model are as follows. The result of LightGBM are lower than those of the deep learning models. By adding new features and optimizing hyperparameter, the performance may be enhanced. For TabNet, we can consider that the feature selection was effective; since the default value was 3 times, we can expect further performance improvement by increasing the number of times and optimizing other hyperparameter. For LSTM, the second highest generalization performance was obtained by considering the time series nature of the data. Further performance improvement can be expected by increasing the number of layers in the model and other adjustments. For Transformer, there is a discrepancy between the cross-validation and the test data, indicating that the model is over-fitting. The cross-validation shows that the accuracy is equivalent to that of TabNet, so eliminating the over-fitting state may result in the highest values.

Table 4: Experimental Results

Models	Cross-validation	Test Data
LightGBM	0.508	0.688
TabNet	0.458	0.545
LSTM	0.474	0.550
Transformer	0.459	0.613

5 CONCLUSION

In this research, we compared the methods for identifying and estimating the factors that cause symptoms in CIPN based on factor analysis and four different machine learning algorithms. In the future, we will work to improve the accuracy of our machine learning models by collecting additional data from in-the-wild experiments and subdividing the data by changing the time-frame.

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